

Keeping Every Object: A Controlled Study of Sketch-and-Text Guided Scene Generation with Diffusion Models

Le Hoang Bao, Thai Ba Cuong, Nguyen Thanh Chuyen

Faculty of Information Technology, Industrial University of Ho Chi Minh City, Vietnam

Manuscript status (5 October 2026). All numbers come from the improved run of 4 October 2026 (code e590c1a, NVIDIA A100), in which every baseline was tuned on the held-out split and every method was scored for FID/KID on the same images; all values, FID/KID included, are final for this run. Tables are generated from the run’s CSV files, so a later run can be swapped in by rebuilding. The pre-registered 8+-object power test (H1/H2) has not been run yet.

Abstract

Sketch-and-text conditioned diffusion models such as ControlNet produce convincing images from a single object sketch, but they lose, miscount and misplace objects as soon as a freehand scene sketch contains several of them. We study this *object-consistency* problem with a controlled protocol and report what does and does not fix it. We build two benchmarks, QuickDraw-Scenes (real freehand object sketches arranged into 1- to 10-object scenes at three abstraction levels) and COCO-Sketch (real photographs), and score object preservation (OPR), class-wise count error, layout IoU and relation accuracy with an open-set detector that no method uses internally, together with a detector ceiling, three-evaluator robustness checks, compute-matched best-of- N controls, one tuned knob per baseline on a held-out split, and paired tests with Holm correction. On a shared Stable Diffusion 1.5 backbone we compare fourteen methods, including OCSD, a two-branch method that generates and selects each object separately, learns identity tokens, composes the scene with blended latents and adds object-aware attention control. Three findings stand. (1) *Object-first composition works*: pasting the separately generated objects onto a ControlNet image and refining with SDEdit (Collage) raises OPR from 65.2% to 80.6% over plain ControlNet on 72 scenes ($p_{Holm} < 0.001$) and ties box-grounded GLIGEN (81.4%), and its variants are the strongest methods in 8- to 10-object scenes (72.4% and 77.6% with best-of-3, vs. 63.5% for GLIGEN and 40.1% for ControlNet). (2) *Once baselines are tuned, the more elaborate OCSD scene stage does not pay off*: OCSD (74.0% OPR) trails tuned GLIGEN (82.2%) and Collage (84.2%), although no gap survives Holm correction; a training-free single-trajectory variant, OCSD-v2, has the best FID (242 vs. 250 for GLIGEN) but localises objects worse. (3) *Negative results*: region-restricted attention, attention-energy guidance and per-scene identity learning do not improve object preservation on the evaluation scenes, even though the tuning split selected them. Rankings are stable across evaluators (Kendall $\tau = 0.67$ – 0.88). We release the benchmarks, code, tuning record and all per-image results.

Keywords: diffusion models, sketch-to-image, scene generation, object consistency, controllable generation, evaluation protocol, negative results.

1. Introduction

Text-to-image diffusion models [1, 2, 3] have made photorealistic synthesis accessible, and spatial adapters such as ControlNet [4] and T2I-Adapter [5] let a user add a sketch to control shape and layout. For a single object this works well. For a *scene* sketch, which a non-expert draws as several rough objects on one canvas, it frequently does not. The control network treats the whole sketch as one geometric map: it knows *where* there are strokes, but not *which strokes belong to which object*, and binding each region to a phrase of the prompt is left to cross-attention. The result is a familiar set of failures: objects disappear (especially small or abstract ones), duplicates merge or multiply, attributes leak between objects, and objects drift away from where they were drawn. Recent analysis shows that scene complexity, not data imbalance, is the main driver of these multi-object failures, and that counting and spatial relations are the least robust abilities [6].

We call the requirement that the generated image contain *the right objects, in the right number, at the right place and shape, in the right relations* **object consistency**. The closest prior work, Zhang et al. [7], splits generation into an object-level stage (generate each object from its own sketch, learn an identity embedding with a masked loss) and a scene-level stage (blend object and background latents early, then denoise freely with identity tokens). We started from that design and extended it into OCSD, adding candidate selection, attention separation and an object-aware conditioning module for the free-denoising phase. A first comparison suggested that OCSD preserved the most objects. When we then tuned one knob per baseline on the same held-out split

used for OCSD and added compute-matched controls, that conclusion did not survive. This paper reports the controlled comparison that replaced it.

This paper makes three contributions:

1. **An evaluation protocol for object consistency** (Section 4). Two automatically built benchmarks with controlled object count and sketch abstraction; detection metrics scored by an evaluator that differs from every detector used inside the methods; a detector ceiling; robustness to the evaluator (stricter IoU, competing queries, a closed-set detector); compute-matched best-of- N controls; baseline tuning on a held-out split with the same rule as the proposed method; and paired tests with Holm correction.
2. **Evidence that object-first composition is what helps** (Section 5.3). Generating and selecting every object separately (our module M2) and then composing them is enough to match a box-grounded model trained for layout and to beat it in crowded scenes, without any per-scene training.
3. **Honest negative results** (Sections 5.2, 5.5, 5.6). With tuned baselines, OCSD’s scene stage, its region-attention and energy-guidance components and per-scene identity learning do not improve object preservation; we quantify how far each is from the best method and show that the tuning split and the evaluation split disagree about them.

2. Related Work

Sketch-to-image generation. Early work used conditional GANs: pix2pix [8] for pixel-aligned edges and SketchyGAN [9] on the Sketchy database [10]. With diffusion models, Voynov et al. [11] guide sampling with a latent edge predictor, and Koley et al. [12, 13] argue that abstract sketches should be interpreted semantically rather than traced pixel by pixel. ControlNet [4], T2I-Adapter [5] and Uni-ControlNet [14] add spatial conditions to a frozen text-to-image model and are the de facto baselines for sketch control.

Scene-level sketches. Sketch2Photo [15] composed retrieved photographs; SketchyCOCO [16] separated foreground and background with EdgeGAN but covers only 14 foreground classes; FS-COCO [17] and SketchyScene [18] provide freehand scene sketches. Unsupervised and text-guided scene sketch-to-photo methods [19, 20] normalise sketches and photos to an edge domain, Cheng et al. [21] address multi-object interference with per-object control, FineControlNet [22] injects region-wise text alongside spatial control, and SketchingReality [23] trains a sketch-semantic modulation network with attention supervision. Zhang et al. [7] is our starting point and is described in Section 1.

Layout and multi-object control. For text prompts, Attend-and-Excite [24] maximises the attention of neglected tokens, Structured Diffusion [25] and SynGen [26] bind attributes through syntax, and Composable Diffusion [27] adds score functions. With boxes, GLIGEN [28] trains gated self-attention layers for grounding, while BoxDiff [29], Layout Guidance [30] and Attention Refocusing [31] steer attention maps with energy functions at inference; DenseDiffusion [32] modulates attention scores with region masks and MultiDiffusion [33] fuses region-wise diffusion paths. Object-level semantic alignment has also been shown to improve multi-object fidelity [34]. These methods use boxes or masks, not the shape information of a freehand sketch, and have no notion of object identity.

Identity and consistency. Textual Inversion [35], DreamBooth [36], LoRA [37] and IP-Adapter [38] personalise a model to a concept; Break-A-Scene [39] extracts several concepts from one image with a masked loss and a cross-attention loss; MasaCtrl, ConsiStory and StoryDiffusion [40, 41, 42] share self-attention to keep subjects consistent across images. OCSD borrows the masked loss and attention supervision for per-scene identity tokens, and combines them with sketch-derived regions.

3. Method

3.1 Problem formulation

The input is a scene sketch S of size $H \times W$ (512×512) containing N foreground objects and a prompt P that contains object phrases $\{p_1, \dots, p_N\}$ (class and attributes, e.g. “two white sheep”) and a background phrase p_{bg} . Each object carries *object information* $o_i = (c_i, m_i, b_i, a_i)$: class, region mask, bounding box and attributes, and the objects are linked by spatial relations $R = \{(i, j, r)\}$ with $r \in \{\text{left, right, above, below, in}$

front, behind}. An output image I is *object-consistent* if (R1) every object appears with the right class and attributes, (R2) the per-class count matches the sketch, (R3) each object occupies its sketched region and follows its shape, and (R4) every relation in R holds. We ask: (Q1) how fast does a sketch-conditioned baseline degrade as object count and sketch abstraction grow; (Q2) does splitting generation into object and scene levels with identity tokens help; and (Q3) does object-aware conditioning reduce the remaining loss, count and relation errors.

3.2 Overview

Figure 1 shows the pipeline. M1 parses the sketch and prompt into objects. The object-level branch generates each object (M2) and learns its identity (M3); the scene-level branch composes the scene with blended and customised inference (M4) under object-aware conditioning (M5). All modules share Stable Diffusion 1.5 [1] with ControlNet Scribble v1.1, so every baseline in Section 5 can run on the same backbone. Section 3.8 describes a training-free single-trajectory variant, and Section 5.1 a simpler object-first baseline (Collage) built from the same M1 and M2.

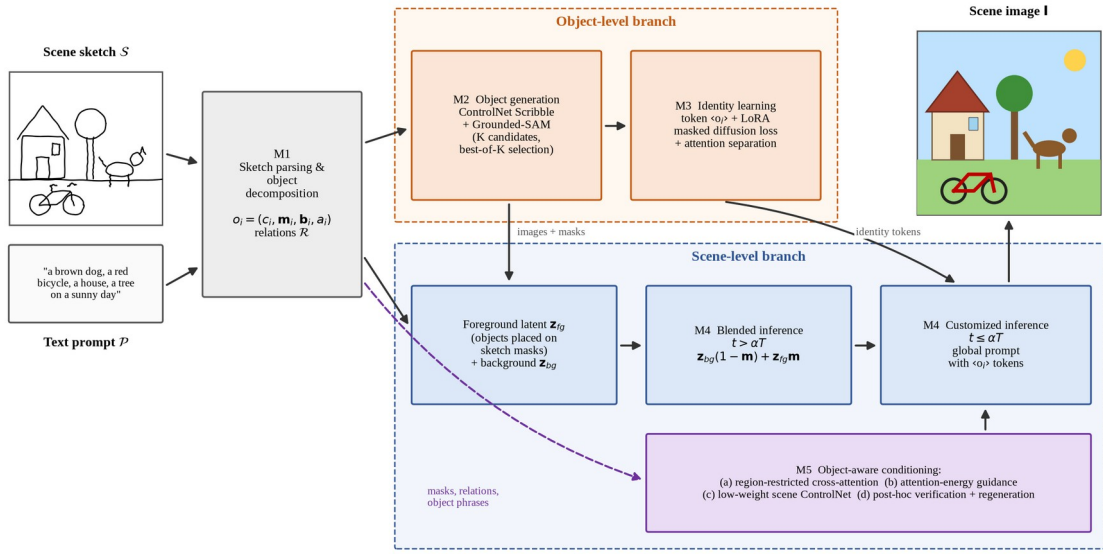


Figure 1. Overview of OCSD. The object-level branch (M2, M3) generates each object from its own sketch and learns an identity token; the scene-level branch (M4, M5) composes the scene with blended inference and then denoises under object-aware conditioning.

3.3 Object decomposition (M1)

The sketch is binarised, padded to 512×512 , thinned and re-thickened to about 3 px to match ControlNet Scribble’s training distribution. Strokes are grouped into objects from drawing layers when the user draws one object per layer (the interactive mode), from benchmark annotations, or, as a fallback, by connected components after dilation. The mask m_i is the convex hull of the object’s strokes after morphological closing, dilated by 9 px, and b_i is its bounding box. Relations follow geometric rules on box centres with a margin $\delta = 0.05W$; “in front of” is decided by the lower box edge under a ground-plane assumption. Object phrases are matched to sketched objects by class and left-to-right order, and the global prompt is rebuilt as $P_g = \langle o_1 \rangle c_1$ and ... $\langle o_N \rangle c_N, p_{bg}$.

3.4 Object-level generation with candidate selection (M2)

Each object sketch is cropped to its box, centred and resized to 512×512 , and K candidates are generated with ControlNet Scribble and the prompt “a photo of a p_i , simple white background”. Using the user’s full phrase, not only the class name as in [7], attaches attributes to the right object from the start. Grounded-SAM [43, 44, 45] segments each candidate, and the candidate with the highest normalised score

$$k^i = \arg \max_k \left[\lambda_1 CLIP(x_i^{(k)} \odot \hat{m}_i^{(k)}, p_i) + \lambda_2 IOU(\hat{m}_i^{(k)}, m_i^{crop}) + \lambda_3 conf_{det} \right]$$

is kept ($\lambda_1 = \lambda_2 = \lambda_3 = 1$), rewarding correct semantics, faithful shape and a confident detection. M2 and M3 run once per scene with a fixed preparation seed and are reused across generation seeds and ablation variants, so that differences between variants come only from the changed component.

3.5 Identity learning (M3)

For each object a new token $\langle o_i \rangle$ is added to the text encoder, initialised from the embedding of c_i , and trained on its object image with the masked diffusion loss

$$L_{id} = \frac{1}{N} \sum_{i=1}^N E_{\epsilon, t} [\| (\epsilon - \epsilon_{\theta, \Delta}(z_{i, t}, t, \text{"a photo of } \langle o_i \rangle c_i \text{"})) \odot \tilde{m}_i \|_2^2].$$

Training has two stages: embeddings only (learning rate 5×10^{-3}), then embeddings (5×10^{-5}) jointly with rank-16 LoRA [37] on the attention projections (10^{-4}). To keep tokens of same-class objects apart, we add an attention-separation term computed on random composites of two or three objects (up to six composites per scene, sampled with probability 0.35):

$$L_{M3} = L_{id} + \lambda_{att} \frac{1}{N} \sum_i \| \hat{A}_i - \tilde{m}_i \|_2^2, \lambda_{att} = 0.01,$$

where $\hat{A}_i$ is the normalised 16×16 cross-attention map of $\langle o_i \rangle$, in the spirit of Break-A-Scene [39].

3.6 Scene composition (M4)

Object images are cut out with their masks, scaled and shifted to maximise IoU with the sketch masks, and pasted far-to-near, giving a foreground image with latent $z_{fg, 0}$ and a union mask $\tilde{m}$. A background latent $z_{bg, t}$ is denoised from noise with the prompt “a photo of p_{bg} ”; its negative prompt lists the foreground classes so that the background does not spawn extra copies of them (applied to both OCSD and our Zhang et al. re-implementation). Denoising then has two phases:

$$z_t \leftarrow z_{bg, t} \odot (1 - \tilde{m}) + z_{fg, t} \odot \tilde{m} (t > \alpha T), z_{t-1} = \text{Denoise}_{\theta, s\Delta}(z_t, P_g, t) (t \leq \alpha T).$$

Blending locks the layout while it is formed; customised inference then harmonises lighting and boundaries while the identity tokens keep each object’s appearance. At inference the LoRA update is scaled by a strength $s \in (0, 1]$. Because z_t is overwritten at every blending step, only the background branch needs to be denoised during blending; the scene U-Net runs from the last blending step on, which gives identical results with about half the U-Net calls.

3.7 Object-aware conditioning (M5)

M5 replaces the plain denoising step of the customised phase with four mechanisms, each aimed at one consistency requirement.

(a) Region-restricted cross-attention (R1, R3). Let T_i index the prompt tokens of object i (its identity token, attributes and class) and T_{bg} those of the background phrase. A bias is added before the softmax,

$$A = \text{softmax}(QK^T / \sqrt{d} + B), B_{n, j} = \begin{cases} -\lambda_t & j \in T_i, n \notin \tilde{m}_i \\ -\lambda_t & j \in T_{bg}, n \in \tilde{m} \lambda_t = \lambda_0 (t / \alpha T)^\gamma, \\ 0 & \text{otherwise,} \end{cases}$$

so each object’s tokens act only inside its own sketch region and the background tokens only outside the foreground; the restriction relaxes towards the end so boundaries can blend. This resembles DenseDiffusion [32], but uses sketch-shaped masks and identity tokens.

(b) Attention-energy guidance (R1, R2). During the first τ customised steps, the latent is updated by $z_t \leftarrow z_t - \eta_t \nabla_{z_t} E$ with

$$E(z_t) = \sum_{i=1}^N \left[\left(1 - \max_{n \in \tilde{m}_i} \dot{A}_i[n] \right) + \beta \frac{\sum_{n \notin \tilde{m}_i} \dot{A}_i[n]}{\sum_n \dot{A}_i[n]} \right].$$

The first term penalises an object that no location attends to strongly (a missing object), as in Attend-and-Excite [24]; the second penalises attention outside the region (drift or duplication), as in BoxDiff and Layout Guidance [29, 30]. Attention maps follow Attend-and-Excite (drop the start token, scale by 100, softmax over text tokens, 3×3 Gaussian smoothing), and η_t decays linearly from η to $\eta/2$. Which tokens the energy acts on is a design choice: steering the identity tokens $\langle o_i \rangle$ turned out to be harmful, and the final configuration, chosen on the tuning split (Section 4.4), steers the object-phrase tokens (attributes and class words) of each object.

(c) Low-weight scene ControlNet (R3, R4). The full scene sketch is fed to ControlNet Scribble with conditioning scale $\omega=0.4$ during customised inference, reminding the model of contours and relative placement after hard blending stops, without imposing strokes on the background.

(d) Post-hoc verification (R1, R2, R4). Grounding DINO (threshold 0.35) detects every sketched class; detections are matched to sketch boxes by same-class Hungarian matching on IoU. An image passes when every object is matched, every class count is correct and every relation holds. Otherwise OCSD regenerates with a new seed and $\lambda_0 \times 1.5$, at most $R=2$ times, and returns the image with the highest $OPR+0.5mIoU+0.25RA-0.25OCE_c/N$. The verifier deliberately differs from the evaluation detector (Section 4.3).

Table 1 lists the hyperparameters. The default column is the full configuration in the code; the experiment column is the reduced setting that fits a Colab Pro budget, with α , s and the M5(b) tokens chosen on a held-out tuning split (Section 4.4).

Table 1. OCSD hyperparameters: code default and the setting used in all experiments.

Symbol	Meaning	Default	Experiments
T, w	DDIM steps, CFG scale	50, 7.5	30, 7.5
K	candidates per object (M2), denoising steps	4, 30	2, 20
S_1, S_2	embedding / joint steps (M3), LoRA rank	200 / 200, 16	100 / 100, 16
α, s	blending switch point, LoRA strength (M4)	0.5, 1.0	0.1, 0.5 (tuned)
λ_0, γ	region-attention strength and decay (M5a)	8, 1	8, 1
β, η, τ	energy guidance (M5b), steered tokens	1, 20, 10 steps, identity	1, 20, 10 steps, phrase (tuned)
ω	scene ControlNet scale (M5c)	0.4	0.4
R	maximum regenerations (M5d)	2	2

3.8 OCSD-v2: a single training-free trajectory

The ablation of the first run suggested that OCSD’s separate background branch and per-scene training might be the weak points, so we also evaluate a training-free variant. OCSD-v2 keeps M1 and M2 but drops M3, and replaces the two-branch sampler of M4 with one joint trajectory: the composed foreground is re-imposed on the scene latent inside the object masks for $t > \alpha T$, and the four M5 mechanisms act from the first step rather than only in the free phase. Energy guidance is averaged over objects instead of summed, very small objects get a minimum mask size in the attention grid, and on COCO the class words of the caption are restricted to the masks of their class. Instead of regenerating the whole image when the M5(d) check fails, OCSD-v2 repairs only the regions of the missing objects. Its α was tuned on the held-out split ($\alpha=0.4$; Section 4.4).

4. Evaluation Protocol

4.1 Benchmarks

Public scene-sketch datasets [16, 17] do not let us vary object count and sketch abstraction independently, and do not provide per-object sketches with phrases. We therefore build two complementary benchmarks, both generated automatically by the released code.

QuickDraw-Scenes. Real freehand object sketches from Quick, Draw! [46] in 28 classes shared with COCO are arranged on a 512×512 canvas with a perspective layout in 12 settings (e.g. “in a green meadow”, “on a city street”), so class, mask, box and relations of every object are known exactly. Scenes form a 4×3 grid: 1, 3, 5 or 8 to 10 objects, times three abstraction levels. Sketch detail is measured as points plus ten times strokes and split into per-class terciles: *simple* uses the most detailed tercile, recognised sketches and no overlap; *medium* the middle tercile with box IoU up to 0.15; *complex* the most abstract tercile, IoU up to 0.30, Gaussian stroke jitter ($\sigma = 2.5$ px) and thinner strokes. To test counting, 40% of objects duplicate an existing class and phrase (“three white sheep”), and the caption states the counts so that text-only methods also know them.

COCO-Sketch. COCO val2017 [47] images with 1 to 8 salient objects (each at least 1.5% of the image and not cut by the border), balanced over 1, 2–3, 4–5 and 6–8 objects, cropped to 512×512 . Sketches are PiDiNet [48] scribbles, split per object by the ground-truth masks; captions are the human COCO captions, which every method receives (for OCSD they are appended to P_g). Real images allow FID/KID and a detector ceiling.

4.2 Protocol

The evaluation uses 72 QuickDraw-Scenes scenes (6 per cell). Methods that need per-scene identity learning (OCSD, Zhang et al.) run on a stratified 36-scene subset (3 per cell) with two seeds; all other methods run on all 72 scenes with seed 0 and additionally with seed 1 on the 36-scene subset, so the main comparison has every method on the same scenes and seeds. On COCO-Sketch, training-free methods run on 32 scenes and all methods on a common 16-scene subset (one seed). The OCSD ablation uses 18 identity-learning scenes with 3 and 5 objects (seed 0 for every row), the OCSD-v2 ablation 18 QuickDraw and 32 COCO scenes, and the α study 12 scenes. Every method uses SD 1.5, DDIM with 30 steps, CFG 7.5, the same negative prompt and the same seeds. Experiments ran on one NVIDIA A100 40 GB in Google Colab.

4.3 Metrics

Detection-based metrics use **OWLv2** [49] (threshold 0.30), queried with the sketched class names; detections are matched to sketch boxes by same-class Hungarian matching with IoU at least 0.1. OWLv2 is deliberately different from Grounding DINO, which OCSD, Collage and the best-of- N baselines use as their internal check, so no method is rewarded for optimising against its own judge.

- **Object preservation rate** $OPR = N_{preserved} / N_{input}$ (R1).
- **Class-wise object count error** $OCE_c = \sum_c |N_{gen,c} - N_{input,c}|$, which, unlike the total count error, cannot cancel a missing object of one class against an extra object of another; and **count accuracy**, the share of images with $OCE_c = 0$ (R2).
- **Layout mIoU** between matched detections and sketch boxes, unmatched objects counting as zero (R3), and **relation accuracy (RA)**, the share of sketch relations preserved (R4).
- **CLIP score** [50] with ViT-L/14 [51] for the whole image, and **object CLIP**, computed on crops at the sketch boxes against “a photo of a p_i ”, which checks attribute binding independently of the detector.
- **ID-Sim**, the DINOv2 [52] cosine similarity between each object region in the scene and its M2 object image (methods that use M2 only), and **FID/KID** [53, 54] against 2,000 COCO val2017 crops, computed for every method on the same 72 QuickDraw-Scenes and 16 COCO-Sketch images.

Detector ceiling and robustness. On COCO-Sketch we score the real photographs themselves: they certainly contain all objects, so their OPR bounds what OWLv2 lets any method reach. To check that conclusions do not depend on the evaluator, OPR is also computed at IoU 0.5, with OWLv2 queried with all classes at once

(competing queries, so a cow cannot be counted as a horse), and with a closed-set COCO-trained DETR; we report Kendall’s τ between rankings.

Fair sampling budget. OCSD may generate up to three times. **ControlNet best-of-3** regenerates the baseline with the *same* Grounding DINO check and selection score, and the **best-of- N** variants of ControlNet and GLIGEN sample up to $N=8$ times with the same check, stopping at the first image that passes; Table 9 reports their measured cost.

Statistics. Metrics are averaged over seeds per scene and then over scenes, reported as mean $\pm$ half-width of a 95% bootstrap interval (2,000 scene resamples) [55]. Methods are compared by two-sided paired Wilcoxon signed-rank tests on per-scene values [56] with Holm correction [57] within each family of comparisons (one reference method against all others, per metric). The tests for the object-first comparisons in Section 5.3 were added after the run and are labelled as such.

4.4 Held-out tuning for every method

All tuning used 16 pilot scenes (12 QuickDraw-Scenes, 4 COCO-Sketch, two seeds) that are **excluded from every evaluation set**, and one rule: maximise OPR + mIoU + RA subject to global CLIP within one point of the reference setting.

OCSD was tuned in three phases. Phase 1 chose $\alpha=0.1$ and LoRA strength $s=0.5$ from $\alpha \in \{0, 0.1, 0.2, 0.3, 0.5\} \times s \in \{0.5, 1.0\}$ (score 1.562 vs. 1.294 for the default $\alpha=0.5, s=1$). Phase 2 compared five energy-guidance options and chose phrase tokens with $\eta=20$ (1.754) over no guidance (1.704) and the original identity-token setting (1.414). Phase 3 crossed M5(a)/M5(b) on/off with $\alpha \in \{0.1, \dots, 0.6\}$ and kept both components with $\alpha=0.1$ (1.774), just ahead of both at $\alpha=0.3$ (1.765) and region attention only (1.762). OCSD-v2 was tuned over $\alpha \in \{0.2, 0.4, 0.6\}$ and chose $\alpha=0.4$ (1.771).

Each baseline had one knob tuned with the same rule: ControlNet scale (1.0 kept), T2I-Adapter scale (0.8), GLIGEN’s grounding weight β (1.0, up from 0.3: OPR 75.2% vs. 66.2% on the tuning split), λ_0 of the region-attention baseline (12), η of the energy baseline (10), the SDEdit strength of Collage (0.2) and α of Zhang et al. (0.1, down from the published 0.5). Variants with best-of-3 or best-of- N inherit their base method’s value. In an earlier version of this study only OCSD was tuned; tuning the baselines changed the conclusions (Section 5.2), which is why we treat it as part of the protocol.

5. Results

5.1 Compared methods

Fourteen methods share the backbone, sampler and seeds. **Sketch-conditioned, single pass:** (1) **SD + ControlNet** Scribble on the whole sketch; (2) **T2I-Adapter** Sketch [5]; (3) **CN + region attention**, DenseDiffusion-style [32] masking with OCSD’s masks and phrases, replacing FineControlNet [22], which has no public SD 1.5 code; (4) **CN + attention energy**, BoxDiff/Attend-and-Excite-style [24, 29]. **Box-grounded:** (5) **GLIGEN** [28] with per-object boxes and phrases (no stroke shape). **Same sampling budget:** (6) **CN best-of-3**, (7) **CN best-of- N** and (8) **GLIGEN best-of- N** . **Object-first:** (9) **Collage**, which pastes OCSD’s M2 objects (same placement as M4) onto a ControlNet image of the whole sketch and harmonises the result with SDEdit [58] under the caption and the scene ControlNet, with no identity learning and no M5(a)/(b); (10) **Collage best-of-3** with the same check; (11) **Zhang et al.** [7], re-implemented in the same code base (one candidate, class-name prompt, no attention separation, no M5); (12) **OCSD-lite**, OCSD without identity learning; (13) **OCSD**; (14) **OCSD-v2**.

5.2 Main comparison: OCSD does not beat the tuned baselines

Table 2. QuickDraw-Scenes, 36 identity-learning scenes $\times$ 2 seeds (mean, with the 95% CI half-width for OPR). FID/KID on the same 72 images per method. Bold: best.

			Count acc. (%)						
Method	OPR (%) ↑	OCE _c ↓	↑	mIoU ↑	RA (%) ↑	Obj. CLIP ↑	ID-Sim ↑	FID ↓	KID×10 ³ ↓
SD + ControlNet	67.4 ± 8.9	2.46	25.0	0.510	36.6	23.52	—	254.9	28.2
T2I-Adapter	65.3 ± 9.2	2.62	29.2	0.453	29.7	23.05	—	254.8	36.0
CN + region attention	68.1 ± 9.2	2.81	29.2	0.507	38.4	23.47	—	258.9	30.4
CN + attention energy	72.9 ± 7.7	2.93	29.2	0.530	42.3	24.25	—	258.8	32.5
CN best-of-3	70.1 ± 8.7	2.47	29.2	0.530	41.1	23.62	—	257.9	29.7
CN best-of-N	71.8 ± 8.4	2.24	30.6	0.551	42.5	23.66	—	260.7	32.3
GLIGEN	82.2 ± 6.4	1.85	45.8	0.669	58.3	24.89	—	249.6	29.0
GLIGEN best-of-N	85.3 ± 5.5	1.26	50.0	0.697	64.5	25.20	—	258.1	38.6
Collage	84.2 ± 6.0	1.46	43.1	0.616	65.5	24.80	0.526	260.6	41.8
Collage best-of-3	86.2 ± 5.2	1.40	50.0	0.640	67.0	24.84	0.538	264.1	47.2
Zhang et al. (re-impl.)	61.6 ± 9.3	2.42	23.6	0.436	35.4	21.65	0.454	272.7	62.1
OCSD-lite	72.2 ± 8.6	2.19	33.3	0.544	46.7	23.74	0.435	283.0	74.8
OCSD	74.0 ± 8.8	2.06	34.7	0.541	50.7	24.00	0.472	280.2	74.3
OCSD-v2	73.2 ± 7.9	3.28	33.3	0.406	45.7	24.53	0.390	242.3	33.2

Tuned GLIGEN and the object-first Collage lead. On QuickDraw-Scenes (Table 2), GLIGEN with a tuned grounding weight preserves 82.2% of sketched objects, Collage 84.2%, and their best-of- N /best-of-3 versions 85.3% and 86.2%. OCSD preserves 74.0%, OCSD-lite 72.2% and OCSD-v2 73.2%, about the level of the strongest attention-control baseline (CN + attention energy, 72.9%) and of CN best-of- N (71.8%). GLIGEN best-of- N has the lowest count error (1.26 vs. 2.06 for OCSD) and the best layout mIoU (0.697 vs. 0.541); Collage best-of-3 has the best relation accuracy (67.0% vs. 50.7%) and identity similarity (0.538 vs. 0.472).

Significance. With Holm correction over the 13 comparisons per metric, OCSD is not significantly different from any method on OPR: it trails GLIGEN by 8.2 points (unadjusted $p=0.13$), GLIGEN best-of- N by 11.3 ($p=0.020$, $p_{Holm}=0.24$), Collage by 10.2 ($p=0.058$) and Collage best-of-3 by 12.2 ($p=0.021$, $p_{Holm}=0.24$), and leads Zhang et al. by 12.4 ($p=0.010$, $p_{Holm}=0.13$). Its layout mIoU is significantly below GLIGEN’s and GLIGEN best-of- N ’s. OCSD-v2 is significantly below GLIGEN best-of- N , Collage and Collage best-of-3 on OPR ($p_{Holm}=0.044$, 0.019 and 0.014) and below OCSD on mIoU (0.406 vs. 0.541, $p_{Holm}<0.01$).

What changed from our earlier run. Before the baselines were tuned, OCSD had the highest OPR (75.4% vs. 72.9% for GLIGEN). OCSD itself barely moved (74.0%), but GLIGEN rose from 72.9% to 82.2% when its grounding weight went from 0.3 to 1.0 ($p<0.05$), and our Zhang et al. re-implementation rose from 44.3% to 61.6% when its blending switch moved from the published $\alpha=0.5$ to 0.1. The earlier ranking was an artefact of comparing a tuned method with untuned baselines, a pitfall that the protocol of Section 4.4 is meant to prevent.

Image quality. With every method scored on the same images, FID is 254.9 for ControlNet, 249.6 for GLIGEN and 280.2 for OCSD (KID $\times 10^3$ 28.2, 29.0 and 74.3). OCSD-v2 has the best FID of all methods (242.3, KID 33.2) and the highest global CLIP among sketch-conditioned methods, so its single trajectory fixes most of the cut-out look of OCSD (Figure 2), but at the cost of localisation. Selection by a detector check costs realism: best-of- N and best-of-3 raise KID for both GLIGEN (29.0 to 38.6) and Collage (41.8 to 47.2).

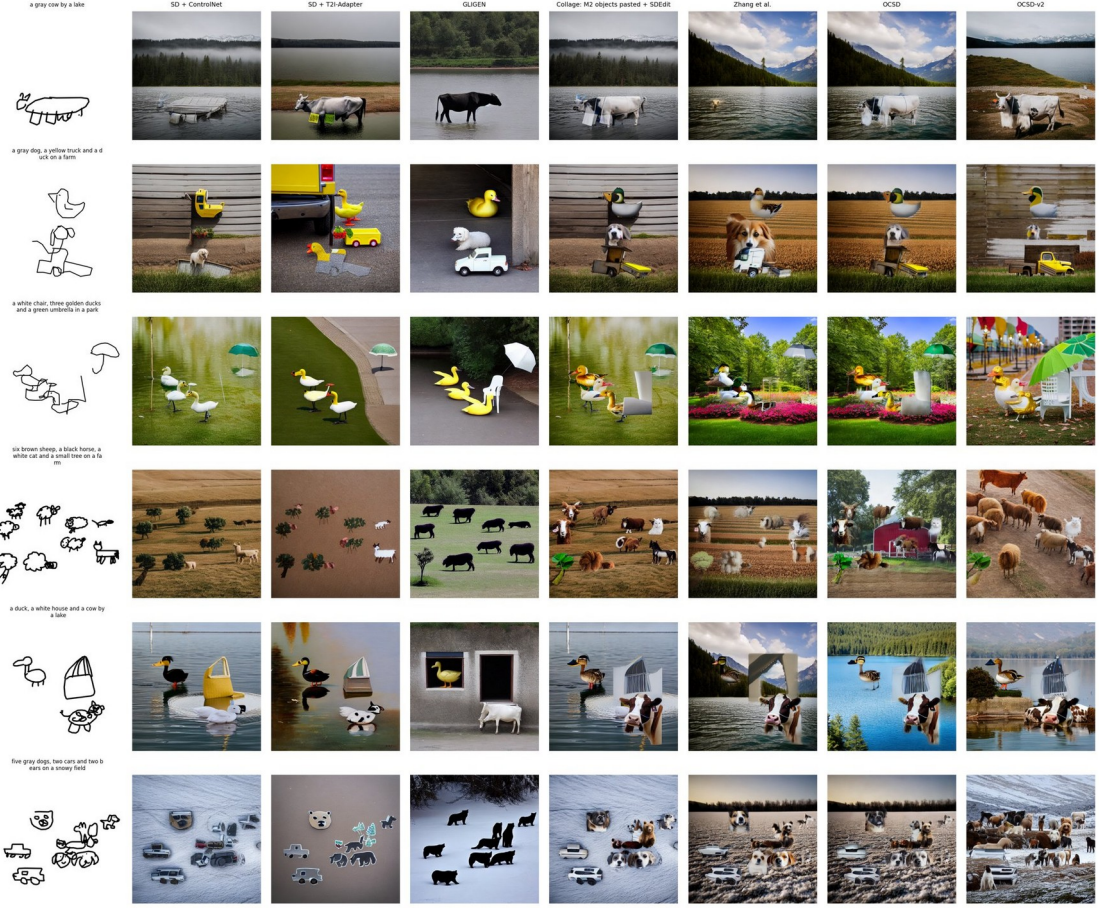


Figure 2. Qualitative comparison on QuickDraw-Scenes (rows chosen automatically). Collage and OCSD keep the sketched objects but show pasted, cut-out objects (rows 1, 2, 5); OCSD-v2 blends them into the scene but drifts from the sketched positions (rows 2, 4). GLIGEN is photorealistic and well placed but ignores stroke shape and attributes (row 4: black instead of brown sheep; row 6: bears instead of dogs).

5.3 Object-first composition is what helps

Table 3. Training-free methods on all 72 QuickDraw-Scenes scenes (every available seed averaged per scene).

Method	OPR (%) $\uparrow$	OCE $\downarrow$	Count acc. (%) $\uparrow$	mIoU $\uparrow$	RA (%) $\uparrow$	CLIP $\uparrow$
SD + ControlNet	65.2 $\pm$ 6.5	2.49	26.4	0.493	32.0	27.98
T2I-Adapter	61.6 $\pm$ 7.2	2.80	29.9	0.428	28.0	28.42
CN + region attention	66.4 $\pm$ 6.8	2.82	29.9	0.489	32.9	27.57
CN + attention energy	71.0 $\pm$ 6.0	2.58	32.6	0.521	39.2	28.51
CN best-of-3	67.8 $\pm$ 6.2	2.35	29.9	0.507	34.6	28.23
CN best-of-N	73.4 $\pm$ 5.9	2.27	36.1	0.564	45.1	28.52
GLIGEN	81.4 $\pm$ 5.2	2.06	47.9	0.662	58.8	28.23
GLIGEN best-of-N	85.5 $\pm$ 4.2	1.49	51.4	0.701	65.8	28.62
Collage	80.6 $\pm$ 5.1	1.59	39.6	0.593	56.9	28.84
Collage best-of-3	81.7 $\pm$ 4.8	1.69	41.7	0.609	58.4	28.74
OCSD-lite	70.7 $\pm$ 6.5	2.17	37.5	0.526	43.5	27.81
OCSD-v2	74.0 $\pm$ 5.9	2.72	34.7	0.419	45.5	28.97

Collage is the simplest way to use separately generated objects: paste them where they were drawn and let a few SDEdit steps harmonise the result. On all 72 scenes (Table 3) it preserves 80.6% of objects against 65.2% for ControlNet (+15.3 points, paired Wilcoxon $p < 0.001$, Holm over five planned comparisons; RA +24.9, mIoU +0.100, both $p_{Holm} \leq 0.001$). Box-grounded GLIGEN gains as much over ControlNet (+16.1 OPR, $p_{Holm} < 0.001$), and Collage and GLIGEN are statistically indistinguishable on OPR (-0.8 , $p = 0.85$) and RA, while GLIGEN places objects more accurately (mIoU +0.069, $p_{Holm} = 0.007$). The same pattern holds for their best-of variants. These planned tests were run after the main analysis (script released with the paper).

The comparison with OCSD-lite isolates the scene stage: both use the same M2 objects and placement, but OCSD-lite composes them with a separate background branch, blended latents and M5, whereas Collage pastes them onto a ControlNet image. Collage preserves 9.8 points more objects ($p_{Holm}=0.012$) and 13.3 points more relations ($p_{Holm}=0.025$). The object-level stage (M2: per-object generation and best-of- K selection) is therefore the part of our pipeline that works, and the elaborate scene stage loses objects that a naive composition keeps.

5.4 Degradation with object count and abstraction

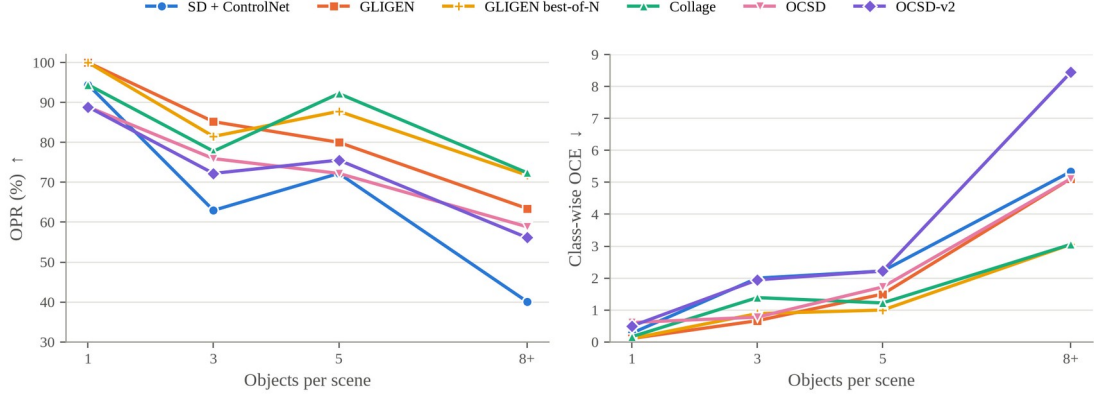


Figure 3. QuickDraw-Scenes, 36-scene subset: OPR (left) and class-wise count error (right) against the number of sketched objects.

Table 4. OPR (%) by object count and by sketch abstraction, QuickDraw-Scenes 36-scene subset.

Method	1 obj.	3 obj.	5 obj.	8+ obj.	simple	medium	complex
SD + ControlNet	94.4	63.0	72.2	40.1	65.6	74.4	62.3
T2I-Adapter	100.0	51.9	66.7	42.6	57.1	75.8	62.9
CN + region attention	94.4	64.8	72.2	40.8	71.7	73.9	58.6
CN + attention energy	94.4	74.1	76.7	46.2	72.2	80.4	66.0
CN best-of-3	94.4	64.8	75.6	45.8	70.8	71.8	67.8
CN best-of-N	94.4	64.8	75.6	52.5	71.0	75.8	68.7
GLIGEN	100.0	85.2	80.0	63.5	81.1	83.9	81.5
GLIGEN best-of-N	100.0	81.5	87.8	71.8	79.4	87.2	89.2
Collage	94.4	77.8	92.2	72.4	85.8	84.0	82.9
Collage best-of-3	100.0	75.9	91.1	77.6	84.8	90.1	83.6
Zhang et al. (re-impl.)	66.7	61.1	66.7	51.9	65.5	60.2	59.0
OCSD-lite	88.9	79.6	62.2	58.1	76.8	73.7	66.2
OCSD	88.9	75.9	72.2	58.9	75.0	80.2	66.8
OCSD-v2	88.9	72.2	75.6	56.2	74.4	77.4	67.9

All methods degrade with object count (Figure 3, Table 4). ControlNet falls from 94.4% OPR on single objects to 40.1% at 8–10 objects, and its count error grows from 0.28 to 5.33. In crowded scenes object-first composition helps most: Collage keeps 72.4% of objects and Collage best-of-3 77.6%, against 63.5% for GLIGEN and 71.8% for GLIGEN best-of- N , with count errors of 3.06 and 2.94 (tied with GLIGEN best-of- N , 3.06) against 5.11 for GLIGEN. OCSD (58.9%) and OCSD-v2 (56.2%) lose this advantage in their scene stage, and OCSD-v2’s count error at 8–10 objects (8.44) shows that its joint trajectory adds extra objects. These 8–10-object figures rest on 9 scenes: one-sided exploratory tests of OCSD against GLIGEN, GLIGEN best-of- N and Collage all give $p_{Holm}=1$, and our pre-registered power test on this regime has not been run. In our earlier, untuned comparison OCSD led GLIGEN here by 16.6 points; that result does not hold against tuned GLIGEN.

Across abstraction levels, GLIGEN, which ignores stroke shape, is stable (81.1 / 83.9 / 81.5% for simple / medium / complex), and GLIGEN best-of- N is best on complex sketches (89.2%). Methods that read the strokes lose most on complex sketches (OCSD 66.8%, ControlNet 62.3%), because ControlNet has to recognise each object from very rough strokes; Collage loses little (82.9%), plausibly because M2’s best-of- K selection discards wrong-class objects before composition.

5.5 Ablations: what does not help

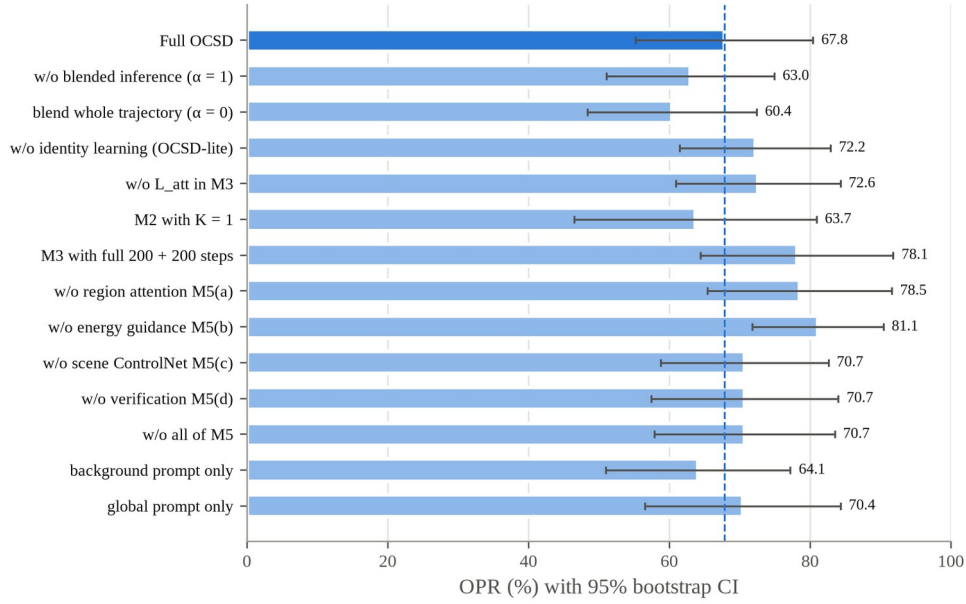


Figure 4. OCSD ablation on 18 QuickDraw-Scenes scenes with 3 and 5 objects (seed 0 for every row): OPR with 95% bootstrap intervals. The dashed line is full OCSD. No difference to full OCSD is significant after Holm correction.

Table 5. OCSD ablation, 18 scenes. Each row changes one component of full OCSD.

Method	OPR (%) $\uparrow$	OCE _c $\downarrow$	mIoU $\uparrow$	RA (%) $\uparrow$	CLIP $\uparrow$	ID-Sim $\uparrow$
OCSD	67.8 $\pm$ 12.6	1.39	0.494	49.4	30.74	0.448
w/o blended inference ($\alpha = 1$)	63.0 $\pm$ 11.9	1.94	0.318	41.3	28.52	0.329
blending over the whole trajectory ($\alpha = 0$)	60.4 $\pm$ 12.0	2.22	0.437	38.8	29.30	0.426
OCSD-lite	72.2 $\pm$ 10.7	1.44	0.534	53.9	30.66	0.412
w/o region attention M5(a)	78.5 $\pm$ 13.1	1.00	0.571	66.6	30.59	0.455
w/o energy guidance M5(b)	81.1 $\pm$ 9.3	1.22	0.571	65.5	31.26	0.485
w/o scene ControlNet M5(c)	70.7 $\pm$ 11.9	1.50	0.504	51.5	31.16	0.459
w/o verification M5(d)	70.7 $\pm$ 13.3	1.28	0.520	54.4	30.38	0.446
w/o all of M5	70.7 $\pm$ 12.8	1.11	0.517	52.7	30.74	0.453
w/o L_{att} in M3	72.6 $\pm$ 11.7	1.61	0.515	55.3	30.73	0.458
M2 with $K = 1$	63.7 $\pm$ 17.2	1.72	0.458	51.9	29.54	0.428
M3 with full 200 + 200 steps	78.1 $\pm$ 13.7	0.78	0.560	66.4	30.98	0.473
background prompt only	64.1 $\pm$ 13.1	2.22	0.449	43.3	29.35	0.401
global prompt only	70.4 $\pm$ 13.9	2.28	0.424	55.3	29.23	0.419

With 18 scenes and one seed, no ablation row differs significantly from full OCSD after Holm correction (smallest $p_{Holm} = 0.24$), so Table 5 shows directions, not established effects. The directions are nevertheless consistent with Section 5.3. The components that touch the *layout* help: removing blended inference ($\alpha = 1$) drops mIoU from 0.494 to 0.318, blending during the whole trajectory ($\alpha = 0$) costs 7.4 OPR points, and M2 with a single candidate costs 4.1 points. The components meant to *enforce* consistency in the free phase do not: removing region attention M5(a) (78.5%) or energy guidance M5(b) (81.1%) scores *higher* than full OCSD (67.8%), as does OCSD-lite without identity learning (72.2%). Training identity tokens twice as long (78.1%) also scores higher, so the reduced M3 schedule may be part of the problem, but none of these variants was tested against Collage on the same 18 scenes.

Table 6. OCSD-v2 ablation on 18 QuickDraw-Scenes scenes (seed 0).

Method	OPR (%) $\uparrow$	OCE _c $\downarrow$	mIoU $\uparrow$	RA (%) $\uparrow$	CLIP $\uparrow$	ID-Sim $\uparrow$
OCSD-v2	76.3 $\pm$ 11.1	2.17	0.410	57.8	31.27	0.387
separate background branch (OCSD sampler)	80.7 $\pm$ 8.2	1.33	0.550	63.4	31.05	0.411
whole-image regeneration instead of repair	73.0 $\pm$ 10.9	2.06	0.391	52.4	30.76	0.351
w/o check and repair M5(d)	71.9 $\pm$ 10.2	2.67	0.368	48.5	31.00	0.353
w/o region attention M5(a)	78.9 $\pm$ 7.4	2.00	0.465	59.5	31.75	0.384
w/o energy guidance M5(b)	75.6 $\pm$ 9.8	1.94	0.451	56.7	30.54	0.408
w/o all of M5	50.4 $\pm$ 11.9	3.11	0.233	26.0	29.50	0.250
objects never re-imposed ($\alpha = 1$)	59.6 $\pm$ 8.3	2.22	0.311	28.8	29.61	0.274
w/o small-object mask fix	75.2 $\pm$ 10.9	2.33	0.406	55.4	31.36	0.381
energy summed over objects	68.5 $\pm$ 11.1	2.00	0.385	44.5	29.83	0.349

The OCSD-v2 ablation (Table 6) gives the same message from the other side. Without any M5 component OCSD-v2 collapses (50.4% OPR), so on a single trajectory some object-aware control is essential, and region repair beats whole-image regeneration (76.3% vs. 73.0%). But replacing the joint trajectory with OCSD’s separate background branch *raises* OPR to 80.7% and mIoU from 0.410 to 0.550, and removing region attention again scores higher (78.9%). On COCO, appending the caption to the prompt lowers OCSD-v2’s OPR from 55.4% to 51.9% (32 scenes, $p_{Holm}=0.26$) while raising CLIP by 2.9 points.

5.6 The tuning split disagrees with the evaluation split

Phase 3 of the tuning (Section 4.4) explicitly compared OCSD with and without M5(a)/(b) on the 16 held-out scenes and kept both (score 1.774 vs. 1.762 with region attention only and 1.644 or less with energy only); the evaluation scenes then preferred dropping either one. Similarly, OCSD-v2 reached 80.3% OPR on the tuning split and 73.2% on the evaluation scenes. With 16 tuning scenes the selection rule cannot resolve differences of this size, so the attention-control components should be read as *not shown to help*, not as shown to hurt. A larger tuning split is the cheapest fix and is part of our next run.

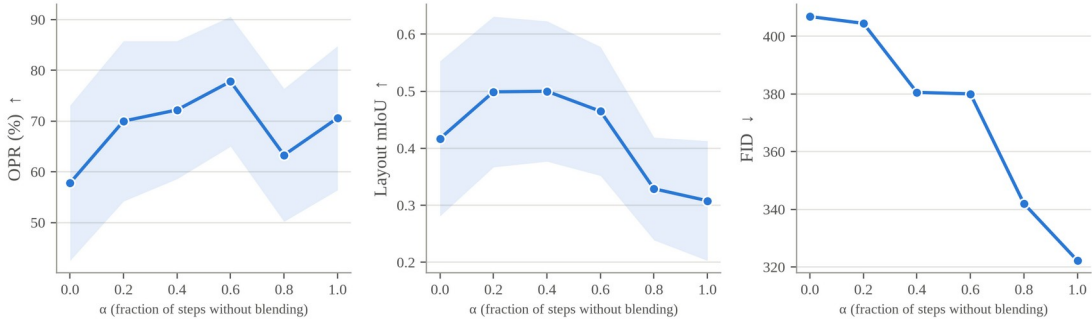


Figure 5. α study on 12 scenes (one seed, verification off): OPR, layout mIoU and FID against α . $\alpha = 0$ blends during the whole trajectory; $\alpha = 1$ never blends.

The blending switch α shows the same trade-off in both runs (Figure 5): layout mIoU is highest for $\alpha \in [0.2, 0.4]$ (0.50) and falls once blending ends early (0.308 at $\alpha = 1$), OPR peaks at $\alpha = 0.6$ (77.8%), and FID improves steadily as the blending phase shortens (406.9 at $\alpha = 0$ to 322.2 at $\alpha = 1$). Layout fidelity and realism pull α in opposite directions, which is the same tension that separates OCSD (good layout, high FID) from OCSD-v2 (worse layout, best FID).

5.7 COCO-Sketch

Table 7. COCO-Sketch, 16 scenes common to all methods (one seed; 95% CI half-widths for OPR are 17–21 points). The real photographs reach OPR 57.7%, OCE_c 2.28, mIoU 0.485 and RA 33.8% under the same evaluator (detector ceiling).

Method	OPR (%)	OCE_c ↓	Count acc. (%)	mIoU ↑	RA (%) ↑	Obj.	ID-Sim ↑	FID ↓	KID×10 ³ ↓
	↑		↑			CLIP ↑			
SD + ControlNet	35.4	2.62	12.5	0.306	10.2	20.67	–	294.9	3.1
T2I-Adapter	36.5	2.44	12.5	0.297	10.4	20.34	–	306.9	5.5
CN + region attention	37.5	2.25	12.5	0.315	14.0	20.97	–	301.3	4.0
CN + attention energy	38.0	2.31	12.5	0.301	13.0	21.13	–	313.1	2.9
CN best-of-3	33.9	2.62	12.5	0.293	9.7	20.96	–	292.6	6.8
CN best-of-N	35.9	2.38	12.5	0.312	12.0	20.88	–	304.8	6.8
GLIGEN	53.5	1.88	31.2	0.458	25.4	22.30	–	282.9	5.6
GLIGEN best-of-N	58.7	1.44	43.8	0.501	37.8	22.12	–	282.6	3.0
Collage	52.5	2.06	12.5	0.414	17.9	22.45	0.509	286.4	12.3
Collage best-of-3	56.6	1.94	18.8	0.452	28.1	22.51	0.518	285.4	13.0
Zhang et al. (re-impl.)	40.0	2.06	12.5	0.313	21.1	21.88	0.417	324.6	37.3
OCSD-lite	52.5	1.88	31.2	0.408	21.1	22.12	0.445	316.1	20.5
OCSD	46.2	2.00	25.0	0.376	21.1	21.99	0.458	315.1	19.0
OCSD-v2	50.4	2.12	18.8	0.361	14.4	21.98	0.434	296.7	8.2

Real COCO scenes are much harder: even the photographs reach only 57.7% OPR under OWLv2, because objects are small and occluded, so every generated result should be read against this ceiling. GLIGEN best-of-N (58.7%) reaches it, Collage best-of-3 (56.6%), GLIGEN (53.5%) and Collage (52.5%) come close, and OCSD (46.2%) and OCSD-v2 (50.4%) fall short; ControlNet reaches 35.4%. With 16 scenes the intervals are about ± 18 OPR points and no difference involving OCSD survives Holm correction, so COCO results are trends. On the 32 training-free scenes, Collage and GLIGEN are again tied on OPR (57.8% vs. 60.7%, $p=0.61$) and both beat ControlNet (42.7%; $p_{Holm}=0.077$ and 0.011). Appending the human caption to OCSD’s prompt raised its CLIP score only from 19.4 to 20.2 and lowered OPR from 53.0% to 46.2% (not significant); OCSD and OCSD-lite remain about 4.5 CLIP points below the single-pass baselines on COCO, which we attribute to PiDiNet scribbles containing background strokes that the object-region constraints fight against.

5.8 Robustness to the evaluator

Table 8. OPR (%) under different evaluators: OWLv2 (default), OWLv2 requiring IoU ≥ 0.5 , OWLv2 with competing class queries, and a closed-set COCO DETR. QD: QuickDraw-Scenes (36 scenes), COCO: COCO-Sketch (16 scenes).

Method	QD OWLv2	QD OWLv2			COCO OWLv2	COCO DETR
		QD OWLv2 @IoU 0.5	+ comp. queries	QD DETR		
SD + ControlNet	67.4	58.8	66.2	59.3	35.4	48.7
T2I-Adapter	65.3	49.9	63.6	54.3	36.5	63.9
GLIGEN	82.2	80.4	79.9	76.7	53.5	94.8
CN + region attention	68.1	58.1	66.1	60.4	37.5	62.5
CN + attention energy	72.9	61.2	72.4	65.7	38.0	57.3
CN best-of-3	70.1	60.5	68.5	65.1	33.9	56.2
CN best-of-N	71.8	63.2	71.1	66.2	35.9	59.7
GLIGEN best-of-N	85.3	83.1	83.6	82.6	58.7	96.9
Collage	84.2	73.0	81.0	73.6	52.5	87.9
Collage best-of-3	86.2	76.5	82.7	76.5	56.6	89.1
Zhang et al. (re-impl.)	61.6	51.7	60.8	60.1	40.0	71.5
OCSD-lite	72.2	66.2	68.9	58.9	52.5	82.8
OCSD	74.0	64.8	70.6	62.5	46.2	82.2
OCSD-v2	73.2	43.5	72.6	72.6	50.4	83.9
Kendall τ vs. OWLv2 ranking	–	–	0.87	0.67	–	0.84

The method ranking is stable across evaluators (Kendall τ with the OWLv2 ranking 0.87 with competing queries and 0.67 with DETR on QuickDraw-Scenes, 0.84 with DETR on COCO-Sketch), and GLIGEN best-of- N , Collage best-of-3, Collage and GLIGEN form the top group under every evaluator. Two differences matter. At IoU ≥ 0.5 , OCSD-v2 drops from 73.2% to 43.5%, against 64.8% for OCSD and 83.1% for GLIGEN best-of- N : its objects exist but are not where they were drawn. And the closed-set DETR is much more lenient on COCO (95.6% for the real photographs vs. 57.7% for OWLv2), so absolute COCO numbers depend strongly on the evaluator while the ranking does not.

5.9 Cost

Table 9. Average time per image on an A100 (s), per-scene preparation for the object stage (M2) and identity learning (M3), and generations per image.

Method	Time / image (s)	M2 / scene (s)	M3 / scene (s)	Generations / image
SD + ControlNet	2.16	–	–	1.0
T2I-Adapter	1.51	–	–	1.0
CN + region attention	2.22	–	–	1.0
CN + attention energy	3.18	–	–	1.0
CN best-of-3	5.21	–	–	2.26
CN best-of- N	11.4	–	–	5.01
GLIGEN	1.89	–	–	1.0
GLIGEN best-of- N	7.53	–	–	3.72
Collage	2.95	12.32	–	1.0
Collage best-of-3	5.95	12.32	–	2.03
Zhang et al. (re-impl.)	1.84	9.04	35.57	1.0
OCSD-lite	4.4	12.41	–	1.92
OCSD	4.87	12.41	35.59	1.9
OCSD-v2	6.86	12.32	–	2.24

ControlNet and GLIGEN take about 2 s per image. Collage takes 3.0 s per image plus 12.3 s of per-scene object generation (M2) that is reused across seeds and prompt edits, which makes it the cheapest method in the top group. OCSD takes 4.9 s per image plus 12.4 s for M2 and 35.6 s for M3; OCSD-v2 avoids M3 but needs 6.9 s per image. The best-of- N baselines are the most expensive per image (7.5 s for GLIGEN, 11.4 s for ControlNet, with 3.7 and 5.0 generations on average).

6. Discussion and Limitations

Answers to the research questions. (Q1) Object consistency of sketch-conditioned diffusion degrades sharply with object count: ControlNet keeps 94% of single objects but 40% at 8–10 objects. (Q2) Splitting generation into object and scene levels helps, but the help comes from the object level: generating and selecting each object separately and pasting it (Collage) lifts OPR by 15 points over ControlNet, ties box-grounded GLIGEN, and is strongest in crowded scenes. (Q3) Object-aware conditioning of the free denoising phase (region attention, energy guidance) and per-scene identity learning do not improve preservation on the evaluation scenes; on a single trajectory (OCSD-v2) some such control is necessary, but the result is still worse than naive composition.

What the evidence does not support. We do not claim that OCSD or OCSD-v2 outperforms the baselines; with tuned baselines both trail GLIGEN and Collage, OCSD-v2 significantly so. The crowded-scene advantage we reported for OCSD in an earlier version disappeared once GLIGEN was tuned. Collage’s advantage in crowded scenes rests on 9 scenes and was not tested by a pre-registered hypothesis.

Lessons for evaluating controllable generation. Three parts of the protocol changed our conclusions and we recommend them generally: tuning one knob per baseline on the same held-out split as the proposed method (it moved GLIGEN by 9 points), compute-matched best-of- N controls with the same internal check (they separate “better sampler” from “more samples”), and a stricter-IoU and alternative-detector check (it exposed OCSD-v2’s localisation problem, which the default OPR hides).

Limitations. All results depend on auxiliary models: Grounded-SAM masks in M2, Grounding DINO in the internal checks and OWLv2 in the evaluation; the COCO ceiling shows that the evaluation detector alone

misses 42% of real objects. QuickDraw-Scenes uses real strokes but synthetic layouts, and COCO-Sketch uses PiDiNet scribbles rather than human scene sketches; FS-COCO [17] is the natural next test set. The experiments use a reduced configuration (30 steps, $K=2$, 100 + 100 training steps) and small samples (36 and 16 scenes, one seed on COCO and in the ablations), so confidence intervals are wide. The tuning split (16 scenes) is too small to rank close configurations reliably (Section 5.6). Collage’s pasted objects are visibly cut out, and its FID (260.6) is worse than GLIGEN’s (249.6). The analyses in Section 5.3 were planned after seeing the main results. A user study, prepared as an anonymised questionnaire following [7], has not been run.

7. Conclusion

We framed sketch-and-text scene generation around object consistency and built an evaluation protocol that measures it with an independent detector, a detector ceiling, robustness checks, compute-matched controls, baseline tuning on a held-out split and paired tests. Under this protocol, our two-branch method OCSD and its training-free variant OCSD-v2 do not outperform tuned baselines: box-grounded GLIGEN and a simple object-first Collage, which reuses OCSD’s per-object generation and selection, preserve 8 to 12 points more objects, and Collage is strongest in crowded scenes. Region-restricted attention, attention-energy guidance and per-scene identity learning did not help on the evaluation scenes. The positive lesson is that generating each object separately and composing it is a strong, cheap and training-free way to keep objects; the methodological lesson is that baseline tuning and compute matching can reverse a ranking. The next step is a method that combines the two strongest ingredients, object-first initialisation and box grounding, evaluated with a pre-registered hypothesis on a larger set. Code, benchmarks, per-image results and the tuning record are available at github.com/bobbibao/ocsd-thesis.

Acknowledgements

The authors thank the Faculty of Information Technology, Industrial University of Ho Chi Minh City, for supporting this work.

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